2.2.1 MECHANICAL DIGESTION

The mechanical processes involved in digestion are:

1. Ingestion refers to intake of food by the mouth. The amount of food taken by an individual is dependent on hunger and appetite. Hunger is the need for food and appetite is the preferential desire for food.
2. Mastication is the process of chewing the food. It ensures that the food taken is broken down into smaller pieces and ground well. The mouth is equipped with teeth which are well designed for chewing. The incisors provide a strong cutting action and the molars a grinding action. All the jaw muscles when working together can close the teeth with a force as great as 55 pounds on the incisors and 200 pounds on the molars.
3. Deglutition is the act of swallowing which involves the voluntary squeezing or rolling action which pushes the food into the pharynx and the involuntary pharyngeal action that pushes the food into oesophagus.
4. Peristalsis is the wave like motion of the esophagus that propels the food into the stomach (Figure 2.9).
5. Mixing and propulsion - Alternating contractions and relaxations of smooth muscle in the walls of the GI tract mix the food with the secretions of the GI tract and propel them forward. This action of the intestine is called as motility.

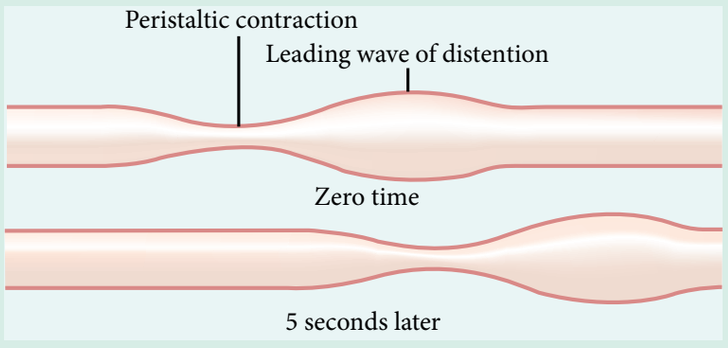


Figure 2.9 Peristaltic Movements

6. Defecation- Digested constituents that were not absorbed in the digestive tract and undigested waste leave the body through the anus in a process called defecation. The eliminated material is termed as feces or stool.

2.2.2 CHEMICAL DIGESTION

2.2.2.1 Digestion and absorption of carbohydrates

The food we eat contains abundant carbohydrates as they provide a major share of daily caloric requirement. Dietary carbohydrates consist of both digestible compounds such as starch, glycogen, lactose, maltose and sucrose and indigestible fibers such as cellulose and hemi-cellulose (Figure 2.10).

Digestion of carbohydrates takes place briefly in the mouth and largely in the n intestine. The polysaccharides get hydrated during cooking, which is essential for efficient digestion.

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